Experiment 4

AIM: Build Online Event Registration Portal using ASP.NET Controls & Validation.

Source: Program4.cs

Code:

index.aspx

```
<%@ Page Title="Registration"
    Language="C#"
    MasterPageFile="~/Site.Master"
    AutoEventWireup="true"
    CodeBehind="Registration.aspx.cs"
    Inherits="RegistrationDemo.Registration" %>

<asp:Content ID="MainContent" ContentPlaceHolderID="MainContent" runat="server">

    <div class="registration-box">

        <h2>Event Registration</h2>

        <asp:ValidationSummary
            ID="ValidationSummary1"
            runat="server"
            HeaderText="Please correct the following errors:"
            ForeColor="Red" />

        <!-- Full Name -->
        <div class="form-row">
            <label>Full Name</label>
```

```
        <asp:TextBox ID="txtName" runat="server"></asp:TextBox>

        <asp:RequiredFieldValidator
            ID="rfvName"
            runat="server"
            ControlToValidate="txtName"
            ErrorMessage="Full Name is required."
            Text="*"
            ForeColor="Red">
        </asp:RequiredFieldValidator>

        <asp:RegularExpressionValidator
            ID="revName"
            runat="server"
            ControlToValidate="txtName"
            ValidationExpression="^[A-Za-z ]{2,50}$"
            ErrorMessage="Name should contain only letters and spaces."
            Text="*"
            ForeColor="Red">
        </asp:RegularExpressionValidator>
    </div>


    <!-- Email -->
    <div class="form-row">
        <label>Email</label>

        <asp:TextBox ID="txtEmail"
            runat="server"
```

```
            TextMode="Email">
        </asp:TextBox>

        <asp:RequiredFieldValidator
            ID="rfvEmail"
            runat="server"
            ControlToValidate="txtEmail"
            ErrorMessage="Email is required."
            Text="*"
            ForeColor="Red">
        </asp:RequiredFieldValidator>

        <asp:RegularExpressionValidator
            ID="revEmail"
            runat="server"
            ControlToValidate="txtEmail"
            ValidationExpression="^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$"
            ErrorMessage="Enter a valid email address."
            Text="*"
            ForeColor="Red">
        </asp:RegularExpressionValidator>
    </div>


    <!-- Mobile -->
    <div class="form-row">
        <label>Mobile Number</label>

        <asp:TextBox ID="txtMobile"
            runat="server"
```

```
            MaxLength="10">
        </asp:TextBox>

        <asp:RequiredFieldValidator
            ID="rfvMobile"
            runat="server"
            ControlToValidate="txtMobile"
            ErrorMessage="Mobile number is required."
            Text="*"
            ForeColor="Red">
        </asp:RequiredFieldValidator>

        <asp:RegularExpressionValidator
            ID="revMobile"
            runat="server"
            ControlToValidate="txtMobile"
            ValidationExpression="^[6-9][0-9]{9}$"
            ErrorMessage="Enter a valid 10-digit mobile number."
            Text="*"
            ForeColor="Red">
        </asp:RegularExpressionValidator>
    </div>


    <!-- College -->
    <div class="form-row">
        <label>College</label>

        <asp:TextBox ID="txtCollege"
            runat="server">
```

```
        </asp:TextBox>

        <asp:RequiredFieldValidator
            ID="rfvCollege"
            runat="server"
            ControlToValidate="txtCollege"
            ErrorMessage="College name is required."
            Text="*"
            ForeColor="Red">
        </asp:RequiredFieldValidator>
    </div>


    <!-- Department -->
    <div class="form-row">
        <label>Department</label>

        <asp:RadioButtonList
            ID="rblDepartment"
            runat="server">

            <asp:ListItem Value="">Select Department</asp:ListItem>
            <asp:ListItem>CSE</asp:ListItem>
            <asp:ListItem>IT</asp:ListItem>
            <asp:ListItem>ECE</asp:ListItem>
            <asp:ListItem>Mechanical</asp:ListItem>
            <asp:ListItem>Civil</asp:ListItem>

        </asp:RadioButtonList>
```

```
        <asp:CustomValidator
            ID="cvDepartment"
            runat="server"
            ErrorMessage="Please select a department."
            Text="*"
            ForeColor="Red"
            OnServerValidate="cvDepartment_ServerValidate">
        </asp:CustomValidator>
    </div>


    <!-- Event -->
    <div class="form-row">
        <label>Event</label>

        <asp:DropDownList
            ID="ddlEvent"
            runat="server">

            <asp:ListItem Text="-- Select an Event --" Value=""></asp:ListItem>
            <asp:ListItem>Hackathon</asp:ListItem>
            <asp:ListItem>Technical Quiz</asp:ListItem>
            <asp:ListItem>Web Development Workshop</asp:ListItem>
            <asp:ListItem>Coding Competition</asp:ListItem>

        </asp:DropDownList>

        <asp:RequiredFieldValidator
            ID="rfvEvent"
            runat="server"
```

```
            ControlToValidate="ddlEvent"
            InitialValue=""
            ErrorMessage="Please select an event."
            Text="*"
            ForeColor="Red">
        </asp:RequiredFieldValidator>
    </div>


    <!-- Gender -->
    <div class="form-row">
        <label>Gender</label>

        <asp:RadioButtonList
            ID="rblGender"
            runat="server">

            <asp:ListItem Value="">Select Gender</asp:ListItem>
            <asp:ListItem>Male</asp:ListItem>
            <asp:ListItem>Female</asp:ListItem>
            <asp:ListItem>Other</asp:ListItem>

        </asp:RadioButtonList>

        <asp:CustomValidator
            ID="cvGender"
            runat="server"
            ErrorMessage="Please select your gender."
            Text="*"
            ForeColor="Red"
```

```
            OnServerValidate="cvGender_ServerValidate">
        </asp:CustomValidator>
    </div>



    <!-- Password -->
    <div class="form-row">
        <label>Password</label>

        <asp:TextBox
            ID="txtPassword"
            runat="server"
            TextMode="Password">
        </asp:TextBox>

        <asp:RequiredFieldValidator
            ID="rfvPassword"
            runat="server"
            ControlToValidate="txtPassword"
            ErrorMessage="Password is required."
            Text="*"
            ForeColor="Red">
        </asp:RequiredFieldValidator>

        <asp:RegularExpressionValidator
            ID="revPassword"
            runat="server"
            ControlToValidate="txtPassword"
            ValidationExpression="^(?=.*[a-z])(?=.*[A-Z])(?=.*\d).{8,}$"
            ErrorMessage="Password must contain at least 8 characters, one uppercase
```

```
letter, one lowercase letter and one number."
                Text="*"
                ForeColor="Red">
            </asp:RegularExpressionValidator>
        </div>



        <!-- Confirm Password -->
        <div class="form-row">
            <label>Confirm Password</label>

            <asp:TextBox
                ID="txtConfirmPassword"
                runat="server"
                TextMode="Password">
            </asp:TextBox>

            <asp:RequiredFieldValidator
                ID="rfvConfirmPassword"
                runat="server"
                ControlToValidate="txtConfirmPassword"
                ErrorMessage="Please confirm your password."
                Text="*"
                ForeColor="Red">
            </asp:RequiredFieldValidator>

            <asp:CompareValidator
                ID="cvPassword"
                runat="server"
                ControlToValidate="txtConfirmPassword"
```

```
            ControlToCompare="txtPassword"
            Operator="Equal"
            Type="String"
            ErrorMessage="Password and Confirm Password must match."
            Text="*"
            ForeColor="Red">
        </asp:CompareValidator>
    </div>


    <!-- Skills -->
    <div class="form-row">
        <label>Skills</label>

        <asp:CheckBoxList
            ID="cblSkills"
            runat="server">

            <asp:ListItem>HTML</asp:ListItem>
            <asp:ListItem>CSS</asp:ListItem>
            <asp:ListItem>C#</asp:ListItem>
            <asp:ListItem>Java</asp:ListItem>
            <asp:ListItem>Python</asp:ListItem>
            <asp:ListItem>JavaScript</asp:ListItem>

        </asp:CheckBoxList>

        <asp:CustomValidator
            ID="cvSkills"
            runat="server"
```

```
            ErrorMessage="Please select at least one skill."
            Text="*"
            ForeColor="Red"
            OnServerValidate="cvSkills_ServerValidate">
        </asp:CustomValidator>
    </div>


    <!-- Address -->
    <div class="form-row">
        <label>Address</label>

        <asp:TextBox
            ID="txtAddress"
            runat="server"
            TextMode="MultiLine">
        </asp:TextBox>

        <asp:RequiredFieldValidator
            ID="rfvAddress"
            runat="server"
            ControlToValidate="txtAddress"
            ErrorMessage="Address is required."
            Text="*"
            ForeColor="Red">
        </asp:RequiredFieldValidator>
    </div>


    <!-- Terms -->
```

```
<div class="form-row">

    <asp:CheckBox
        ID="chkTerms"
        runat="server"
        Text=" I agree to the Terms & Conditions" />

    <asp:CustomValidator
        ID="cvTerms"
        runat="server"
        ErrorMessage="You must agree to the Terms & Conditions."
        Text="*"
        ForeColor="Red"
        OnServerValidate="cvTerms_ServerValidate">
    </asp:CustomValidator>

</div>


<!-- Register Button -->
<div class="form-row">

    <asp:Button
        ID="btnRegister"
        runat="server"
        Text="Register"
        CssClass="register-button"
        OnClick="btnRegister_Click" />

</div>
```

```
        <!-- Success Message -->
        <div class="message">

            <asp:Label
                ID="lblMessage"
                runat="server">
            </asp:Label>

        </div>

    </div>

</asp:Content>
```

Registration.aspx.cs

```
using System;
using System.Text;
using System.Web.UI.WebControls;

namespace RegistrationDemo
{
    public partial class Registration : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
        }

        // Department validation
```

```
protected void cvDepartment_ServerValidate(
    object source,
    ServerValidateEventArgs args)
{
    args.IsValid = !string.IsNullOrEmpty(rblDepartment.SelectedValue);
}


// Gender validation
protected void cvGender_ServerValidate(
    object source,
    ServerValidateEventArgs args)
{
    args.IsValid = !string.IsNullOrEmpty(rblGender.SelectedValue);
}


// Skills validation
protected void cvSkills_ServerValidate(
    object source,
    ServerValidateEventArgs args)
{
    bool skillSelected = false;

    foreach (ListItem skill in cblSkills.Items)
    {
        if (skill.Selected)
        {
            skillSelected = true;
            break;
```

```
            }
        }

        args.IsValid = skillSelected;
    }


    // Terms and Conditions validation
    protected void cvTerms_ServerValidate(
        object source,
        ServerValidateEventArgs args)
    {
        args.IsValid = chkTerms.Checked;
    }


    // Register button
    protected void btnRegister_Click(object sender, EventArgs e)
    {
        // Check all validation controls
        if (!Page.IsValid)
        {
            lblMessage.Text = "";
            return;
        }

        // Collect selected skills
        StringBuilder selectedSkills = new StringBuilder();

        foreach (ListItem skill in cblSkills.Items)
```

```
            {
                if (skill.Selected)
                {
                    if (selectedSkills.Length > 0)
                    {
                        selectedSkills.Append(", ");
                    }

                    selectedSkills.Append(skill.Text);
                }
            }

            // Display successful registration
            lblMessage.Text =
                "Registration successful!<br/><br/>" +
                "Name: " + Server.HtmlEncode(txtName.Text) + "<br/>" +
                "Email: " + Server.HtmlEncode(txtEmail.Text) + "<br/>" +
                "Mobile: " + Server.HtmlEncode(txtMobile.Text) + "<br/>" +
                "College: " + Server.HtmlEncode(txtCollege.Text) + "<br/>" +
                "Department: " + Server.HtmlEncode(rblDepartment.SelectedValue) + "<br/>"
+
                "Event: " + Server.HtmlEncode(ddlEvent.SelectedValue) + "<br/>" +
                "Gender: " + Server.HtmlEncode(rblGender.SelectedValue) + "<br/>" +
                "Skills: " + Server.HtmlEncode(selectedSkills.ToString()) + "<br/>" +
                "Address: " + Server.HtmlEncode(txtAddress.Text);
        }
    }
}
```

Site.Master

```
<%@ Master Language="C#" AutoEventWireup="true"
  CodeBehind="Site.Master.cs"
  Inherits="RegistrationDemo.SiteMaster" %>

<!DOCTYPE html>

<html>
<head runat="server">
  <title>Event Registration</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f5f5f5;
      margin: 0;
      padding: 30px;
    }

    .registration-box {
      width: 500px;
      margin: 0 auto;
      background-color: white;
      padding: 25px;
      border: 1px solid #ddd;
      border-radius: 6px;
    }

    h2 {
      margin-top: 0;
      text-align: center;
    }

    .form-row {
      margin-bottom: 14px;
    }

    .form-row label {
      display: block;
      margin-bottom: 5px;
      font-weight: bold;
    }

    input[type="text"],
    input[type="email"],
    textarea,
    select {
      width: 100%;
      box-sizing: border-box;
```

```
        padding: 7px;
      }

      textarea {
        height: 80px;
      }

      .register-button {
        padding: 8px 20px;
        cursor: pointer;
      }

      .message {
        margin-top: 15px;
        font-weight: bold;
      }
    </style>
</head>

<body>
    <form id="form1" runat="server">

      <asp:ContentPlaceHolder ID="MainContent" runat="server">
      </asp:ContentPlaceHolder>

    </form>
</body>
</html>
```

Program4.cs

```
namespace RegistrationDemo
{

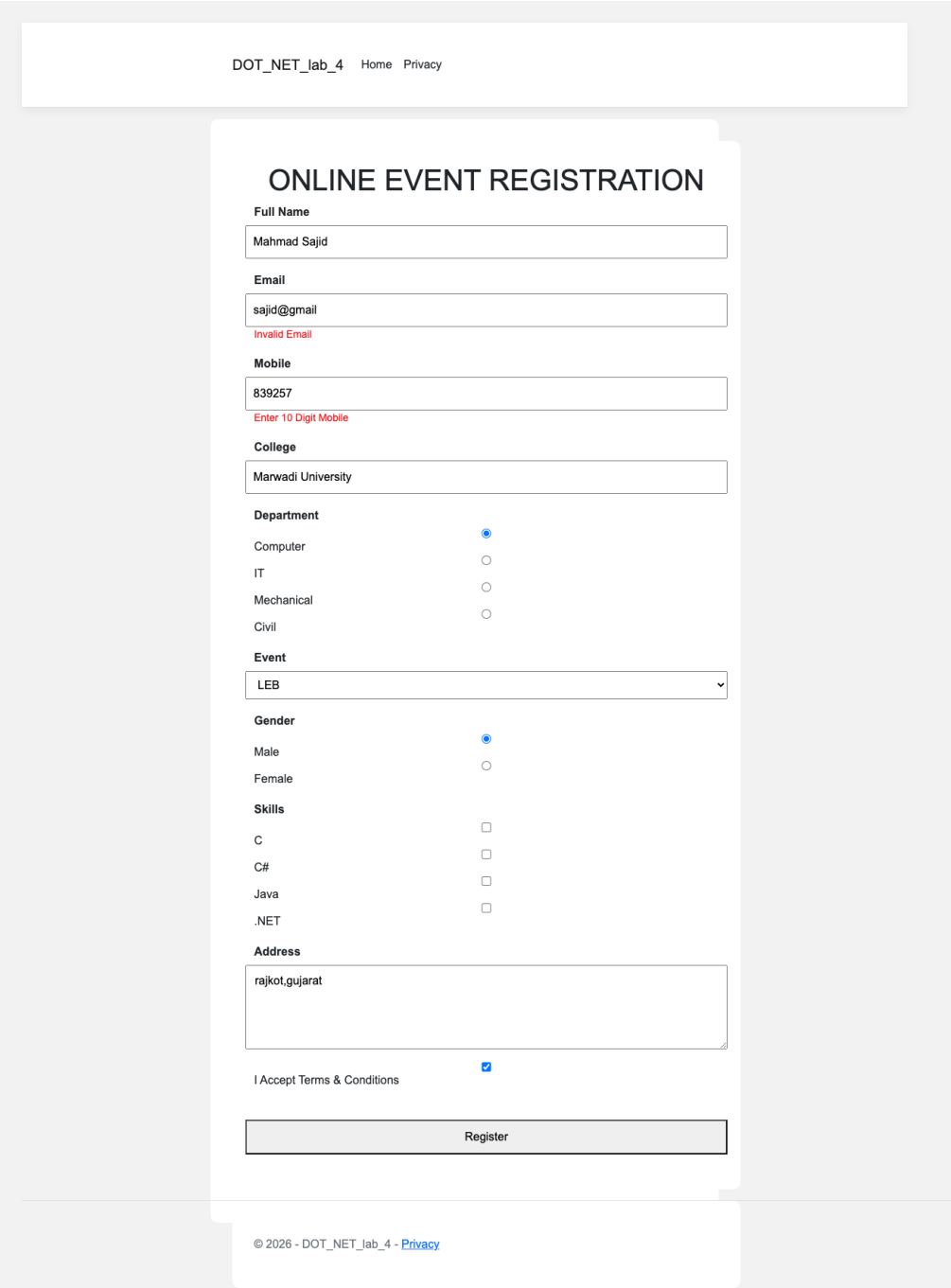

    public partial class SiteMaster
    {


        protected global::System.Web.UI.HtmlControls.HtmlForm form1;

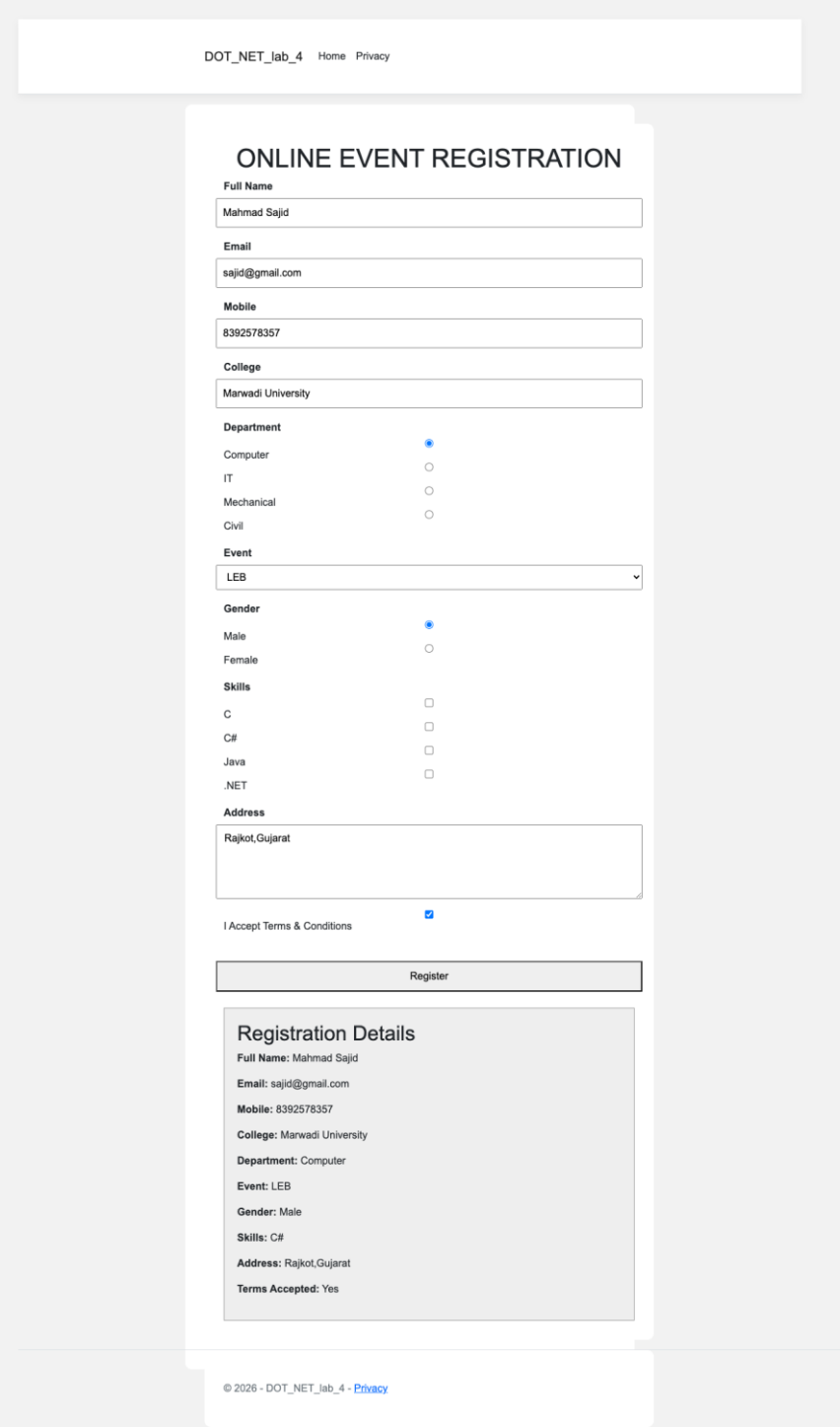

        protected global::System.Web.UI.WebControls.ContentPlaceHolder MainContent;
    }
}
```

Output Screenshot:

Registration Form violating different validation conditions:



Registration Form Successfully Submitted:



GitHub URL: https://github.com/Ashish-Agrawal-7322/.Net.git